Which rules of thumb hold at his level?

The app's rules of thumb, followed as players at his level would follow them, on 128,021 Lichess positions refereed by Stockfish

Chess Coach research lab

Abstract

A human coach explains a middlegame or endgame move by a rule of thumb: walk the king toward the pawns, put the rook behind the passed pawn, take the open file. The app teaches fifteen such rules, and its board code says when a move shows one. In rated 10-minute Lichess games between players at his level and the band above, the move a rule points to, as a player at that level would play it, is sound less often than the move such a player would naturally play, for all 13 rules with enough positions, on the train months and on the held-out month: by 2.1 points for the rule of the square up to 49.5 for the pawn break. That verdict, fixed before the run, cannot by itself tell a rule from a random choice, so a control added after it draws in each position as many random legal moves as the rule had and keeps the one Maia-2 likes best. That move trails the natural move too, by 16.8 to 65.0 points, and against it twelve rules point to better moves than chance on both periods; only cutting the king off does not. The rules carry information, then, but following one where it differs from the natural move costs 6.5 to 55.8 points. In 104 held-out examples, 38 of the 56 unsound rule moves allow mate or lose material by the app's own blunder check, and 15 more pass over a check, a capture or a promotion that the natural move took. A rule of thumb is a reason to look at a move once the checks, captures and threats are seen, not a reason to play it.

1 The question

A coach explains a move by a rule of thumb more often than by a line: the king is a fighting piece once the queens are off, a rook belongs behind a passed pawn, a knight is strongest on a square no pawn can attack. The app teaches these rules as concepts, and its move detectors say when a move shows one. That is the kind of understanding a player can carry from one game to the next, so it is worth knowing which rules hold at his level.

In the middlegames and endgames that players at his level reach, how often is the move a rule points to sound, is it sounder than the move a player at his level would naturally play, and where does the rule fail? A rule that holds is one the app can teach as a way to choose a move. One that does no better than the natural move is not worth a lesson for that, and one that does worse misleads.

2 How it was measured

The games. Rated standard Lichess games at 10 minutes with no increment, his time control, 30,000 a month in file order: July and August 2026 to train on and September 2026 held out, 90,000 games with 4,294,043 middlegame and endgame positions in all, the phase read by the app's divider. A game enters when both players are in his band or the band above: his band is 1250 to 1449 on

Lichess, around his chess.com rapid rating by the app's offset, and the band above 1450 to 1649. On Maia-2's scale they read 1300 and 1500.

The rules. Fifteen, each read by the app's own code at commit `f4bd63f19b`. A move shows a rule when the app's move description holds a finding of that concept for the mover with the tone of a gain. The outside passed pawn, the Lucena position and the Philidor position are only ever written as notes, so their notes count, each for the side the position is for; trading when ahead is read from its technique reader, which the move description could not reach (a gap reported to the app and since fixed); and the worst piece is read from the app's activity reading: a move by the piece the app names the mover's worst, after which it has more legal moves. Tactics and mates are left out: they are calculation, which the puzzles train.

The positions. For each rule, a game gives at most one position, drawn at random among its positions where the rule is available (some legal move shows it), up to 4,000 a month for each rule: 128,021 positions in all.

The moves compared. In each position, R is the rule's move: among the legal moves that show the rule, the one Maia-2 rates most likely at the mover's band, the rule as a player at that band would play it. N, the natural move, is Maia-2's most likely move of all, and A the move played.

The referee. Stockfish through the app's engine service at its grading budget of 250,000 nodes, the hash cleared before each position: the best is searched with one line, and each move is scored at the same budget. A move is *sound* when it gives up no more than 3 win% against the best. With seven men or fewer the app's tablebase probe decides: a move is sound when it keeps the mover's outcome.

The rule, fixed before any position was scored. The *lift* is R's soundness minus N's, paired by position, with 95% intervals from 2,000 bootstrap draws over games (seed 20261018). A rule is taught when the lift's interval lies above 0 on the train months and on the held-out month; it misleads when it lies below 0 on both; it is no better than the natural move otherwise; and under 300 held-out positions there is not enough data. Reported beside the verdict, a rule is a *step up* when the band above plays R more often than his band, the difference's interval above 0 on both periods. A split of the positions, read before the move, is a *failure condition* when in one of its cells R is sound at least 10 points less often than in the rest, with the difference's interval below 0, on both periods.

3 The verdicts: every rule misleads

Every rule with enough data misleads (Table 1). On the held-out month the rule's move is sound less often than the natural move by 2.1 points for the rule of the square up to 49.5 for the pawn break, and the train months give the same verdicts. The rule's move is sound less often than the move actually played as well, for every rule. The Lucena and Philidor positions came up 6 and 30 times in the held-out month, too few for a verdict.

Only the outpost is a step up: players in the band above play its move 3.1 points more often than players in his band (0.6 to 5.6 on the held-out month). How findable the rules' moves are varies widely: at his band Maia-2 gives the rule's move 6% on average for the rarest and 62% for the most common.

	Held-out positions	R sound	N sound	A sound	Lift, train	Lift, held out	Verdict	A step up
Active king	4,000	57.8	82.9	78.7	−25.2	−25.1 (−26.8, −23.5)	It misleads	no
Opposition	1,435	82.3	93.3	91.4	−9.3	−11.0 (−12.9, −9.1)	It misleads	no
Rule of the square	2,582	96.1	98.2	96.9	−2.0	−2.1 (−2.8, −1.4)	It misleads	no
Rook behind the passed pawn	4,000	35.3	77.5	71.4	−42.3	−42.2 (−43.9, −40.6)	It misleads	no
Cutting the king off	1,398	42.4	83.8	78.9	−41.7	−41.3 (−44.1, −38.6)	It misleads	no
Outside passed pawn	1,303	67.8	84.8	81.3	−16.6	−17.0 (−19.2, −14.9)	It misleads	no
Trading when ahead	4,000	71.4	84.2	76.3	−11.9	−12.8 (−14.1, −11.4)	It misleads	no
Lucena position	6	100.0	100.0	100.0	0.0	0.0 (0.0, 0.0)	not enough data	no
Philidor position	30	86.7	93.3	96.7	−9.6	−6.7 (−20.0, 6.7)	not enough data	no
Open file	4,000	42.7	74.1	67.8	−31.8	−31.4 (−33.2, −29.6)	It misleads	no
Rook on the seventh	4,000	47.0	76.8	70.3	−29.0	−29.8 (−31.5, −28.0)	It misleads	no
Outpost	4,000	43.3	75.0	67.0	−32.2	−31.6 (−33.3, −29.9)	It misleads	yes
Pawn break	4,000	24.8	74.4	64.9	−48.9	−49.5 (−51.3, −47.8)	It misleads	no
Passed pawn	4,000	44.3	77.9	70.6	−33.1	−33.7 (−35.3, −31.9)	It misleads	no
Worst piece	4,000	34.6	75.5	67.6	−40.3	−40.8 (−42.6, −39.1)	It misleads	no

Table 1: The preregistered verdicts. R, N and A sound: the share of held-out positions where the rule's move, the natural move and the move played are sound, in percent; the lift is R's soundness minus N's in points, held out with its 95% interval from 2,000 bootstrap draws over games (for every judged rule the train interval lies below 0 as well). A step up: the band above plays R more often than his band, on both periods.

4 A rule with no information

The run's review found that the lift's zero is not a neutral rule's. R is the move Maia-2 likes best among the moves that show the rule and N the one it likes best of all, so R is N wherever N shows the rule, and wherever R differs, Maia-2 rated it lower. Any subset of the legal moves would trail N, or at best tie it, by this measure. So a control, written down before it was computed, draws in each scored position as many legal moves at random as the rule had, without replacement and seeded by the position, and keeps the one Maia-2 likes best, R_0, refereed as R was. The *random rule's lift*, R_0's soundness minus N's, is what a rule with no information shows; the *rule's content*, R's minus R_0's, is what the rule adds to chance. It changes no verdict.

The random rule misleads too, and by more: its held-out lift lies 16.8 to 65.0 points below 0 (Table 2). Against it, twelve rules point to better moves than chance on both periods, from 2.7 points for the worst piece to 52.2 for trading when ahead. Only cutting the king off reads no differently from chance: its content is 0.3 points (−1.7 to 2.1) on the train months and 0.4 (−2.5 to 3.1) held out. The random moves took 102,887 fresh Stockfish searches, and each found the best the run had stored.

So the rules know something: the moves they point to are sound more often than moves drawn at random in the same number. But wherever the rule's move is not the natural move it is sound less often, by 6.5 points for the rule of the square up to 55.8 for the outside passed pawn on the held-out month, for every rule on both periods. How often the two agree varies just as widely, from 6% of the held-out positions for the pawn break to 69% for the outside passed pawn.

	Random rule's lift, held out	Content, train	Content, held out	Reading	R is N, held out	Lift where R is not N, held out
Active king	−32.1	7.6 (6.5, 8.6)	7.0 (5.5, 8.5)	better	23	−32.4 (−34.4, −30.4)
Opposition	−18.0	8.9 (7.4, 10.4)	7.0 (4.9, 9.0)	better	33	−16.4 (−19.1, −13.7)
Rule of the square	−16.8	14.8 (13.8, 15.8)	14.6 (13.1, 16.2)	better	67	−6.5 (−8.7, −4.4)
Rook behind the passed pawn	−48.0	5.7 (4.7, 6.7)	5.8 (4.3, 7.2)	better	7	−45.4 (−47.2, −43.8)
Cutting the king off	−41.7	0.3 (−1.7, 2.1)	0.4 (−2.5, 3.1)	no different	6	−44.1 (−46.9, −41.1)
Outside passed pawn	−45.5	27.1 (24.9, 29.5)	28.5 (25.4, 31.7)	better	69	−55.8 (−61.1, −50.4)
Trading when ahead	−65.0	51.6 (50.3, 52.7)	52.2 (50.5, 54.0)	better	62	−33.7 (−36.9, −30.3)
Open file	−42.5	11.5 (10.4, 12.7)	11.1 (9.4, 12.7)	better	16	−37.4 (−39.4, −35.3)
Rook on the seventh	−52.5	23.4 (22.3, 24.6)	22.8 (21.2, 24.4)	better	21	−37.6 (−39.6, −35.5)
Outpost	−53.6	20.0 (18.8, 21.2)	21.9 (20.3, 23.6)	better	25	−42.0 (−44.1, −39.8)
Pawn break	−54.1	3.7 (2.7, 4.7)	4.6 (3.1, 6.1)	better	6	−52.4 (−54.3, −50.5)
Passed pawn	−48.8	12.5 (11.4, 13.7)	15.2 (13.6, 16.8)	better	23	−44.0 (−45.9, −41.8)
Worst piece	−43.5	5.5 (4.3, 6.6)	2.7 (1.1, 4.3)	better	11	−45.9 (−47.8, −44.0)

Table 2: The control added after the run. The random rule's lift is the soundness of R_0 minus N's, whose interval lies below 0 on both periods for every rule; the rule's content is R's minus R_0's, in points with 95% intervals from the same 2,000 draws over games. Reading: better than chance when the content's interval lies above 0 on both periods, no different otherwise. R is N: the share of held-out positions where the rule's move is the natural move, in percent.

5 Where the rules fail

The failure conditions were fixed with the rule (Table 3). The rule's move fails when it is quiet for nine rules, when the material is level for nine, when the opponent has a threat (a mate in one or a winning capture, by the app's reading of the position) for eight, in the middlegame for six, and when the mover's king shows a danger sign for two. With a threat on the board the gap is 14 to 30 points: the active king's move is sound 37% of the time when the opponent threatens something and 67% when not. No rule fails where there is no threat or where its move is a capture or a check, and the rule of the square, the outside passed pawn and trading when ahead have no failure condition at all.

Level material is probably in part the referee's doing, an inference this study does not test: a fixed window in win% is narrower in centipawns where the game is balanced than where it is won or lost, so fewer moves pass it.

Rule	Failure conditions: the cell, and R's soundness there minus the rest, held out
Active king	they threaten (−30), material level (−18), R quiet (−42)
Opposition	they threaten (−14), material level (−13), king safe (−18)
Rule of the square	none
Rook behind the passed pawn	they threaten (−22), material level (−20), middlegame (−22), R quiet (−41), king in danger (−12)
Cutting the king off	R quiet (−26)
Outside passed pawn	none
Trading when ahead	none
Open file	they threaten (−15), material level (−20), middlegame (−19), R quiet (−35)
Rook on the seventh	they threaten (−21), material level (−17), R quiet (−20), king in danger (−16)
Outpost	material level (−16), middlegame (−16), R quiet (−24)
Pawn break	they threaten (−18), material level (−15), middlegame (−17), R quiet (−47)
Passed pawn	they threaten (−20), material level (−16), middlegame (−22), R quiet (−16)
Worst piece	they threaten (−18), material level (−19), middlegame (−21), R quiet (−15)

Table 3: Where the rule's move fails: a cell of a split, read before the move, where R is sound at least 10 points less often than in the rest of the split, with the difference's 95% interval below 0, on the train months and the held-out month. The difference shown is the held-out one, in points.

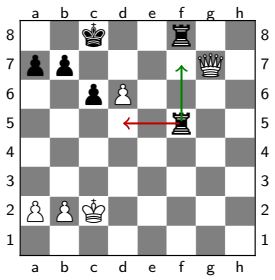
6 What the examples show

The control added eight held-out positions for each judged rule, drawn at random among those where the rule's move is not the natural move: 104 in all, 91 refereed by Stockfish and 13 by the tablebase (`results/examples.md`). After the results, each was read with the app's own blunder check, the reading its live coach makes before it warns about a move (Table 4, `results/why.md`): whether a move allows mate in one, and whether it loses material, either as a capture the capture war says loses or by leaving a piece that can be taken for a gain, newly loose or loose already. The capturing piece of an even trade is not counted, since taking it back is the trade.

Of the 56 rule moves the referee found unsound, the check flags 38: two allow mate and 36 lose material. Of the rest, 15 pass over a check, a capture or a promotion that the natural move took, and only three were a quieter choice that was simply worse. The check flags six of the 48 sound rule moves too, and the natural move 11 times in 104. The natural move is a check, a capture or a promotion in 53 of the examples, the rule's move in 22. Figure 1 shows one rule move of each of the first three kinds, chosen to show them; every example is in the results.

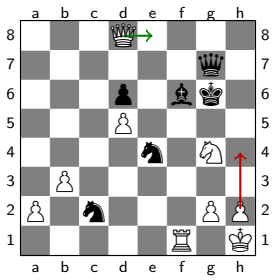
	R sound	Allows mate	Loses material	Passes over a forcing move	A quieter choice	N forcing	Flagged: R / N
Active king	5	0	0	3	0	4	0 / 0
Opposition	5	0	1	1	1	2	2 / 0
Rule of the square	8	0	0	0	0	4	1 / 1
Rook behind the passed pawn	4	1	3	0	0	3	5 / 2
Cutting the king off	2	0	5	1	0	4	7 / 1
Outside passed pawn	2	0	5	1	0	6	5 / 0
Trading when ahead	4	0	1	2	1	5	1 / 0
Open file	4	0	2	2	0	4	3 / 0
Rook on the seventh	4	0	4	0	0	5	4 / 1
Outpost	2	0	4	2	0	4	4 / 0
Pawn break	3	0	4	1	0	3	4 / 4
Passed pawn	1	1	5	1	0	5	6 / 1
Worst piece	4	0	2	1	1	4	2 / 1
All	48	2	36	15	3	53	44 / 11

Table 4: The 104 held-out examples, 8 a rule drawn at random where R is not N, read after the results by the app's blunder check. Each unsound rule move is counted once, under the first kind that fits in the column order. N forcing: the natural move is a check, a capture or a promotion. Flagged: the moves the check says allow mate or lose material, sound or not.



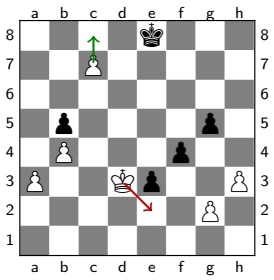
`2k2r2/pp4Q1/2pP4/5r2/8/8/PPK5/8 b - - 7 42`

(a) Black to move. The rook goes behind White's passed d-pawn with ♖d5 (red), and ♕c7# follows. The best, ♖5f7 (green), is the natural move: it closes the seventh rank.



`3Q4/6q1/3p1bk1/3P4/4n1N1/1P6/P1n3PP/5R1K w - - 2 38`

(b) White to move, the queen on d8 attacked by the bishop on f6. The rule pushes the passed h-pawn, h4 (red), and leaves her there: 3% to win, against 98% after the natural move and best, ♕e8+ (green), which moves her with check.



`4k3/2P5/8/1p4p1/1P3p2/P2Kp2P/6P1/8 w - - 1 49`

(c) White to move. The king takes the opposition with ♔e2 (red), 50% to win, while the c-pawn can queen with check: c8=♕+ (green), the natural move and best, 96%.

Figure 1: Three held-out positions where the rule's move is unsound: it allows mate, it leaves the queen to be taken, and it passes over a promotion. Arrows: the rule's move in red, the referee's best in green; win% is the mover's, by Stockfish at 250,000 nodes.

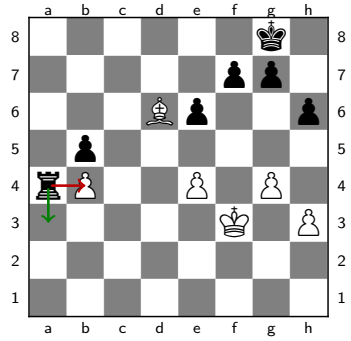
7 Do the detectors read too widely?

The study tests each rule as the app's board code reads it, so a misleading verdict may be the detector's fault. In the examples most detectors read their rule as it is taught: the king walks toward the pawns, the rook goes behind a passed pawn or to the seventh rank, a knight or bishop lands on a square no pawn can attack, a pawn strikes a chain. What fails is not the reading but the moment, as above. Two read wider than a coach would (Figure 2).

The outside passed pawn is read from a note the app writes on any move after which the mover has a new passed pawn far from the other pawns and from their king. The note does not ask what the move gave for it. All eight of its examples' rule moves are captures, and five of them lose material by the capture war, each a rook for a pawn (Figure 2a). Its verdict is in part the reading's, though its moves are still sound far more often than random ones. The app wrote the same note on such a move in Game Review, beside the warning that the rook is lost; since chess PR 166, after the commit measured here, the review holds a technique note back on a move that gives its piece away.

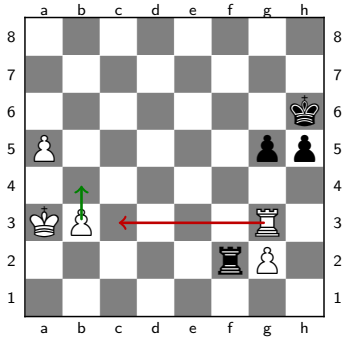
Cutting the king off is read on any rook move, in a rook ending, to a file with nothing else on it between their king and one of the mover's passed pawns. It does not ask whether their king could reach the pawn without the cut, and it is the one rule whose moves are no better than chance (Figure 2b).

The worst piece and the open file are each available in most positions (63% and 66% of the held-out ones): nearly any rook move to a file without the mover's pawns shows the open file, and the worst piece, the lab's own reading rather than the app's, takes any move that gives the least mobile piece more moves. Such a rule rarely says when; the worst piece carries the least content of the twelve.



`6k1/5pp1/3Bp2p/1p6/rP2P1P1/5K1P/8/8 b - - 1 32`

(a) Black to move. ♖xb4 (red) takes the pawn and leaves Black an outside passed b-pawn, which the note names, but the bishop takes the rook back: the capture war counts 4 points lost. The best, ♖a3+ (green), is the natural move.



`8/8/7k/P5pp/8/KP4R1/5rP1/8 w - - 0 37`

(b) White to move. ♖c3 (red) puts the rook on the empty c-file between the black king on h6 and White's queenside pawns, which that king was nowhere near, and leaves g2 to the black rook. The best, b4 (green), pushes the b-pawn.

Figure 2: Two readings wider than a coach's, from the held-out examples. Arrows as in Figure 1.

8 What it means for the app

A rule of thumb is a reason to look at a move and a way to explain one; at his level it is not a way to choose one. The rules point to better moves than chance, so they belong in the app's teaching. But the move a rule points to, played as his level would play it, is sound less often than the move his level plays naturally, and where the two differ the rule's move mostly ignores something forcing: a piece left loose, a mate in one allowed, a check, capture or promotion passed over. The app's own blunder check catches most of these.

The app already uses its rules that way in Game Review: it praises a rule only on a move that gave up less than its praise bar, and names a rule the player missed only on Stockfish's best move. Its pawn-break lesson now says to check what a break leaves hanging before playing it (chess PR 130, after the commit measured here), and its worst-piece lesson begins "No tactics?". Its open-file and outpost lessons taught their rule with no such check, and the evidence here says they should start from it; since chess PR 165 they do, looking at every check, capture and threat first and treating the open file or the hole as where to look once those are answered.

Two readings were for the review to weigh. The outside passed pawn's note was a fact about the board written whatever the move cost, so it appeared beside a blunder as if it were the move's reason; the review now holds it back on a move that gives its piece away (chess PR 166). And the cut-off reading carries no information at this level; the app's endgame drill for it is a position where the cut is the answer, which this study does not test.

The next round, rules-after-check, tested the coach's advice as a coach gives it: the rule's move among the moves the blunder check passes, in positions with no tactic on the board, against the natural move among the same moves. Its control failed, so it gives no verdict, but there too the rule's move is sound less often than the natural move, for every rule with enough positions.

9 Caveats

- **Not his games.** The positions are Lichess players' in his band and the band above, by the app's conversion of his rating; his own games are too few for most rules.
- **One move for a plan.** A rule is tested as the next move his level would play for it, not as a plan over several moves.
- **Sound is not instructive.** A sound move can be the second-best plan; the lift compares the rule's move only with the natural move.
- **A strong baseline.** Maia-2's most likely move at the mover's band is sound more often than the move actually played, for every rule (Table 1).
- **Added afterwards.** The control, the examples and their reading by the blunder check were added after the rule was applied. They describe and change no verdict, and the examples are few, eight a rule.